

PROJECT_INSTRUCTIONS.md — QB-TimeWarp



READ FIRST: `/home/ubuntu/memory.md`

Before doing ANYTHING else, read `/home/ubuntu/memory.md` .

That is the ONE AND ONLY memory file (not in this project dir).

It contains all passwords, connection details, paths, current status, and the complete quick-start checklist for immediate session resumption.



ALSO READ THESE SUPPLEMENTAL FILES

Before starting work, check for and read these files if they exist:

Always Read

File	Purpose
<code>/home/ubuntu/memory.md</code>	🧠 THE memory file — passwords, paths, status, next steps
<code>PROJECT_CONTEXT.md</code>	What the project does, design decisions, requirements
<code>CHANGE_LOG.md</code>	Full chronological history of every change made

Supplemental Files (ALL in `/home/ubuntu/`, NOT the project dir!)

```
# Run this EVERY session to read supplemental files:
cat /home/ubuntu/memory.md
cat /home/ubuntu/PROJECT_INSTRUCTIONS_ADDITION.md 2>/dev/null
cat /home/ubuntu/DEBUGGING_NOTES.md 2>/dev/null
cat /home/ubuntu/CHANGE_LOG_ADDITION.md 2>/dev/null
cat /home/ubuntu/DEBUGGING_SUMMARY_FOR_JOSEPH.md 2>/dev/null
cat /home/ubuntu/debugging_review_report.md 2>/dev/null
```

File	Location	Purpose
memory.md	/home/ubuntu/	THE canonical memory file — all credentials, paths, status
PRO-JECT_INSTRUCTIONS_ADDITION.md	/home/ubuntu/	Root cause analysis of import regression — 2 bugs, QBXML requirements, fix priorities
DEBUGGING_NOTES.md	/home/ubuntu/	Deep debugging: success detection bug, QBXML parsing errors, code architecture notes
CHANGE_LOG_ADDITION.md	/home/ubuntu/	Regression timeline: what broke in Turn 45, 359 phantom successes discovery
DEBUGGING_SUMMARY_FOR_JOSEPH.md	/home/ubuntu/	Concise debugging summary formatted for Joseph
debugging_review_report.md	/home/ubuntu/	Debugging review report

Legacy Files in Project Dir (still valid context)

File	Purpose
PROJECT_INSTRUCTIONS.MD (uppercase)	Original agent briefing from earlier session
DEBUGGING_NOTES.MD (uppercase)	Debugging notes copy in project dir

Reference As Needed

File	Purpose
<code>PATH_REFERENCE.md</code>	All file paths + old→new rename mapping
<code>ARCHITECTURE.md</code>	Code structure, service layers, data flow
<code>TESTING_CHECKLIST.md</code>	Step-by-step testing procedures
<code>POPUP_HANDLING.md</code>	How to handle QuickBooks popups
<code>SAFETY_FEATURES.md</code>	Multi-layer protection system docs
<code>SESSION_RESUME.md</code>	Connection details (also in <code>memory.md</code>)
<code>README_FOR_NEW_SESSIONS.md</code>	Quick-start guide
<code>PASSWORDS.txt</code>	All credentials (gitignored)

PROJECT GOAL

Migrate data from **QuickBooks 2023** (SDK 16.0) to **QuickBooks 2021** (SDK 15.0) via a .NET 6.0 C# application that:

1. **Exports** all entity data from QB 2023 as JSON
2. **Transforms** the data for QB 2021 compatibility (field mapping, format preservation, SDK version adaptation)
3. **Imports** the transformed data into QB 2021 using staged, dependency-aware import
4. **Validates** the migration (entity counts, field comparison, financial reconciliation, format checks)

Key Constraints

- QBXML SDK is 32-bit only → build must target **x86**
- QuickBooks must be running with company file open for SDK to connect
- **Single-user mode** recommended for reliable SDK access
- Originals are NEVER modified — working copy system protects them

CRITICAL BUILD NOTES

```
# dotnet is NOT in PATH – always use full path:  
/home/ubuntu/.dotnet/dotnet build  
  
# Always build the PROJECT, never look for a .sln:  
cd /home/ubuntu/QB-TimeWarp  
/home/ubuntu/.dotnet/dotnet build    # uses QB-TimeWarp.csproj automatically  
  
# Expected result: 0 Errors, 1 Warning (CS8602 in QBConnectionManager.cs:385)
```

WORKING WITH JOSEPH

- Joseph is a **40-year programmer** and MSP owner — he knows what he’s talking about
- He watches the VM screen and programs alongside you — this is **collaborative**
- He wants **precise, complete documentation** with no shortcuts or placeholders
- **Don’t barrel through** — pause at key stages and coordinate
- Keep QB windows **VISIBLE** — never minimize them during SDK operations
- Check `POPOP_HANDLING.md` for QuickBooks dialog handling procedures